RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Sunday, 16 August 2026, 10:35 PM
State	Finished
Completed on	Sunday, 16 August 2026, 10:35 PM
Time taken	14 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Returns index of first occurrence of 0, or m if no 0 exists
int findFirstZero(int arr[], int low, int high) {
    if (low > high) {
        return high + 1;   // no zero found in this range
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == 0) {
        // Check if it's the FIRST zero (either mid==0 or previous element is 1)
        if (mid == low || arr[mid - 1] == 1) {
            return mid;
        } else {
            // there's a zero before mid too, search left half
            return findFirstZero(arr, low, mid - 1);
        }
    } else {
        // arr[mid] == 1, first zero (if any) is in the right half
        return findFirstZero(arr, mid + 1, high);
    }
}

int main() {
    int m;
    scanf("%d", &m);

    int arr[m];
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    int firstZeroIndex = findFirstZero(arr, 0, m - 1);
    int countZeros = m - firstZeroIndex;

    printf("%d\n", countZeros);

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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